26.10.15 (E).

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2015

Max. Marks: 100 | **Duration: 03Hrs**
Class: SE | Semester: III
Name of the Course: Object Oriented Programming Methodology | Branch: Computer Engg.
Course Code: **UCEC302**

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Explain data types in JAVA	05
Q 1 (b)	What is command line argument? Write a program to find the largest of three integers accepted from command line.	10
Q2 (a)	Create a class it has following data members:- name account id balance initialize the data members through constructors. Define a print function & define a function to compare the name. it returns true if the name is equal or else returns false. Initialize an array of objects. Pass a name from the command line & search for the name in the array of objects. if it is existing print the information of the object. **OR** Create an appropriate Class myDate which has two constructors, one default and another parameterized constructor. Create another class arraySwap which accepts two input dates from user in format dd/mm/yy. Write a swap( ) method and use Array of Objects to swap the two dates .Use toString( ) method to display the output in dd/mm/yy format.	10
Q2 (b)	Write a program that uses Vector class to add five names of student initially. Display a menu: (i) Add new name (ii) Delete name (iii)Display name	10
Q.2 (c)	Write the difference between Abstract class and final class	05

Q3 (a)	Write a Program to accept a string from the user. Use a Buffer Reader To accept the input and count the number of words in the string without using inbuilt function of String class. **OR** Write a Program to check whether given string is palindrome or not	10
Q3 (b)	1) Compare String and StringBuffer. 2) Compare Vector and Array.	05 05

Q4 (a)	interface Mat
	
	{
	
	void read( );
	
	void display( );
	
	}
	
	Create a class Matrix by implementing interface Mat. Derive class MatrixOp from Matrix and provide functions to add and multiply two matrices. Also derive class MatrixSearch from Matrix and add a function 'search ( )' which searches given value in the matrix.
		10

Q4 (b)	Draw class Diagram for the College Management System. Clearly show attributes multiplicities and aggregations/compositions/Association between classes in the class diagram.	10
Q5 (a)	Explain with a neat diagram life cycle of thread and Explain any five methods of thread class. **OR** Write a program which accepts marks of a student (between 0 to 100) and checks whether it is within the range or not. If it is within the range then it displays "marks entered successfully", if not then it throws the exception of user defined class "MarksOutOfRangeException". The class should contain appropriate toString method to describe object the class with the out of range marks entered by the user.	10
Q5 (b)	Explain applet life cycle and write an applet program to display Smiley face. **OR** Write a program to display **(1A2B3C......)** using multiple threads.(First thread to display integer and second thread to display Alphabet)	10